

Rajat M

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Professional Summary

- **Machine Learning & Software Engineer** with a Bachelor of Engineering in IT from Mumbai University and hands-on experience developing, training, and deploying ML and Deep Learning models.
- Proficient in handling **large datasets (300k+ records, 120+ features)**, performing feature engineering, exploratory data analysis, class imbalance handling, and data preprocessing.
- Experienced in building and optimizing **Convolutional Neural Networks (CNNs)**, transfer learning pipelines (Xception), object detection (YOLO), and time-series predictive modeling.
- Hands-on experience in productionizing models using **Python, SQL, Docker, Linux environments**, and JSON pipelines.

Technical Skills

- **Programming & Querying:** Python (Advanced), SQL, JavaScript, Bash Shell Scripting
- **Machine Learning & Deep Learning:** Scikit-Learn, TensorFlow, Keras, CNNs, Transfer Learning, YOLO, Feature Engineering, Model Evaluation (Precision/Recall/ROC-AUC)
- **Data Analytics & Data Science:** Pandas, NumPy, SciPy, Matplotlib, Exploratory Data Analysis (EDA), Time-Series Preprocessing
- **Deployment & Infrastructure:** Docker, OpenVINO, Linux/Unix, Git, GitHub, JSON Logging,
- **Computer Science Fundamentals:** Data Structures, Algorithms, System Design, Object-Oriented Programming

Professional Experience

Machine Learning Engineer — BitVivid Solutions

Feb 2026 – Present

- Engineered end-to-end video analytics and object detection pipelines by training and fine-tuning YOLO models on custom datasets.
- Executed full lifecycle ML workflows: raw data collection, automated annotation pipelines, preprocessing, spatial data augmentation, and model optimization.
- Built containerized inference modules using **Python, OpenVINO, and Docker**, optimizing real-time frame processing latency and throughput.
- Integrated automated JSON logging and event snapshot triggers for production deployment across edge and server environments.

Key Projects

Home Loan Default Prediction Using ML | [Project Report & Architecture](#)

Engineered an end-to-end risk classification pipeline on a large dataset of 300,000+ records and 120+ features to predict financial default probability.

- Performed extensive missing-value imputation, outlier treatment, feature scaling, and categorical encoding.
- Handled severe class imbalance using advanced resampling methods and evaluated models using ROC-AUC, Precision, and Recall.
- **Tech Stack:** Python, Pandas, NumPy, Scikit-Learn, SQL, Matplotlib

Deep Learning Fashion Image Classification | [Project Report & Architecture](#)

Architected a multiclass deep neural network pipeline utilizing transfer learning (Xception) on 3,800+ images across 10 categories.

- Applied dynamic data augmentation (rotation, zoom, flips) to prevent overfitting and improve generalization performance.
- Evaluated model convergence using confusion matrices, F1-scores, and training loss metrics in TensorFlow/Keras.
- **Tech Stack:** Python, TensorFlow, Keras, OpenCV, CNNs

Systematic Equity Reversion Backtesting Engine | [Project Report & Architecture](#)

Developed a 10-year walk-forward predictive time-series pipeline on Indian equity market constituents.

- Implemented 24-month rolling out-of-sample validation splits to eliminate look-ahead bias and curve-fitting.
- Evaluated signal quality using Spearman Rank Information Coefficient (IC) and integrated per-trade transaction costs.
- **Tech Stack:** Python, Pandas, NumPy, Time-Series Analysis

Education

Bachelor of Engineering in Information Technology

University of Mumbai | Mumbai, India

Relevant Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Data Mining, Database Management Systems (SQL), Cloud Computing, Distributed Systems.